

System Monitor (Sysmon) Audit Documentation

Practical Lab Report: Installation, Process Tracking & Network Event Analysis

ANALYST	TARGET SYSTEM	DATE	STATUS
Godfred Acheampong	DESKTOP-U77E2R8	August 21, 2026	COMPLETED

Executive Summary

This technical lab document details the setup, configuration, and validation of System Monitor (Sysmon v15.21) for endpoint security visibility on Windows. The primary objective was implementing host-level event auditing, specifically capturing process creation telemetry, operational log driver status, and real-time outbound network connection monitoring.

1. Lab Deliverables & Verification Matrix

Artifact File Name	Target Event	Description & Telemetry Verified	Status
01-sysmon-event-id-1.png	Event ID 1 (Process Creation)	Logged process executions including ProcessGuid, ProcessId, Command Line string, User context, and Parent Image binary path.	VERIFIED
02-sysmon-service-state.png	Service Operational State	Validated active Sysmon service/driver state, baseline schema installation, and active log pipeline in Event Viewer.	VERIFIED
03-sysmon-event-id-3.png	Event ID 3 (Network Connection)	Logged outbound network connections generated by chrome.exe and curl.exe, capturing IP addressing, protocol (UDP), and port bindings.	VERIFIED

2. Technical Execution & Step-by-Step Procedure

Phase 1: Binary Directory Navigation

Navigated to the extracted Sysmon directory in Administrator Command Prompt to issue administrative driver flags:

```
cd "C:\Users\Godfred\Downloads\Sysmon (2)"
```

Phase 2: Sysmon Driver Configuration Reset & Network Logging

To bypass suppression rules and force network connection tracking across the system, the schema configuration was reset and network capturing re-enabled:

```
# Reset active configuration to default schema
sysmon -c -

# Enable schema-wide network connection logging (-n)
sysmon -c -n
```

Driver Configuration Output:

- Network connection: enabled
- DNS lookup: enabled
- HashingAlgorithms: SHA256
- No rules installed (All network events un-filtered)

Phase 3: Network Event Triggering & Verification

Generated outbound web traffic to populate Event ID 3 telemetry in Microsoft-Windows-Sysmon/Operational:

```
curl.exe https://www.google.com -o NUL
```

Web browsing via Google Chrome concurrently logged multiple Event ID 3 instances under protocol UDP (Port 443 / QUIC) and DNS lookups.

3. Captured Telemetry Properties (Event ID 3)

The captured **Event Properties - Event 3, Sysmon** dialog box confirmed the following metadata fields:

- **Event ID:** 3 (Network connection detected)
- **Image:** C:\Program Files\Google\Chrome\Application\chrome.exe
- **User Context:** DESKTOP-U77E2R8\Godfred
- **Protocol:** udp
- **Source Computer:** DESKTOP-U77E2R8
- **Log Stream:** Microsoft-Windows-Sysmon/Operational